

# Ibrutinib and venetoclax

Ibrutinib and venetoclax is a cancer drug treatment. It is used to treat chronic lymphocytic leukaemia (CLL) and small lymphocytic lymphoma (SLL).

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## What is ibrutinib and venetoclax?

This treatment is a combination of the targeted therapy drugs [ibrutinib](#) and [venetoclax](#). It is used to treat [chronic lymphocytic leukaemia \(CLL\)](#) and [small lymphocytic lymphoma \(SLL\)](#). It may sometimes be used to treat other cancers.

It is best to read this information with our general information about [targeted therapy drugs](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

## More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

## How ibrutinib and venetoclax is given

Ibrutinib and venetoclax both come in tablets. This means you can take them at home.

During a course of treatment, you will meet someone from your cancer team, such as a:

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- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

## **Your course of treatment**

Your cancer team will discuss your treatment plan with you. They may give you a copy of the treatment plan to take home.

They will give you the tablets to take home. They may give you tablets of different strengths. Always take the tablets exactly as your cancer team tell you to. This is important to make sure they work as well as possible for you.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

During this treatment, you should avoid grapefruit, star fruit (carambola) and Seville oranges. This is because they can affect how ibrutinib and venetoclax work. You should avoid:

- drinking the juice of these fruits
- eating food that contains them
- taking supplements that might contain them.

## **During week 1 to 12 of this treatment you take ibrutinib only**

For the first 12 weeks of this treatment, you take ibrutinib only.

You take ibrutinib once a day. Try and take the tablets at the same time each day. Swallow the tablets whole with a glass of water. Do not chew, crush or break them.

If you forget a dose of ibrutinib, take the tablets as soon as possible on the same day. Take the next dose at the usual time the next day. Never take a double dose to make up for the one you missed.

Your doctor, nurse or pharmacist may ask you to stop taking ibrutinib or take a lower dose because of side effects. Always follow their advice.

## **From week 13 you also start taking venetoclax**

From week 13 you take ibrutinib and you take venetoclax.

Venetoclax can cause a side effect called tumour lysis syndrome (TLS). There is more information about possible symptoms of TLS below.

To reduce the risk of TLS, you start on a low dose of venetoclax and the dose is gradually increased. During this time, your cancer team will monitor you closely with blood tests after each dose.

You usually take venetoclax in the morning then have the blood tests in a day unit or clinic as an outpatient. Some people may need to stay in hospital during this time. Your cancer team will tell you when your blood will be checked.

Your cancer team will also give you more information about the following ways to prevent TLS:

- drinking plenty of water 2 days before you start venetoclax and every day while the dose is increased
- having extra fluids through a drip when you start venetoclax or when the dose is increased
- taking tablets called allopurinol or having a drip with a drug called rasburicase. These drugs reduce the amount of uric acid in the blood. This is a waste product that causes TLS.

You take venetoclax tablets once a day. Take them with or after food. Swallow them whole with a glass of water. Do not chew, break or crush them. Take them at the same time every day.

If you forget a dose of venetoclax, you can take the missed dose up to 8 hours late. If you miss a dose by more than 8 hours, take the next dose at the usual time. Never take a double dose.

Your doctor, nurse or pharmacist may ask you to stop taking venetoclax because of side effects. Always follow their advice.

## **Other things to remember about your tablets**

- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused tablets. Return them to your cancer team at the hospital.

## **About side effects**

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

## **Serious and life-threatening side effects**

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

## Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

## Common side effects

### Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

**It is important to follow any specific advice your cancer treatment team gives you.**

You will have regular blood tests during treatment. If your white blood cell count is low, your doctor may stop your treatment for a short time, until your cell count increases. Sometimes they will reduce the dose of your treatment.

## Bruising and bleeding

Ibrutinib can increase the risk of bleeding. It can reduce the number of platelets in your blood or affect how they work. Platelets are cells that help the blood to clot.

Tell your doctor, nurse or pharmacist if you are taking any medicines that may affect bleeding. This includes aspirin, blood-thinning tablets such as warfarin, or injections such as heparin or vitamin E.

Bruising on your arms is quite common with ibrutinib. But if you develop large bruises or unexplained bleeding during your treatment, contact the hospital straight away on the 24-hour number. This includes:

- nosebleeds
- bleeding gums
- heavy periods or unexpected vaginal bleeding
- blood in your urine (pee) or stools (poo)
- vomiting or coughing up blood
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

You may need a drip to give you extra platelets. This is called a platelet transfusion.

If you need to have surgery during treatment, your doctor may ask you to stop taking ibrutinib for a few days before and after your operation.

## **Anaemia (low number of red blood cells)**

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

## **Tumour lysis syndrome (TLS)**

This treatment can cause a condition called tumour lysis syndrome (TLS). When treatment makes large numbers of cancer cells die and break down quickly, they release lots of waste products into the blood. This can affect how well the kidneys work. It may also cause problems with the heart rhythm.

The risk of TLS is highest when you start taking venetoclax. You will have regular blood tests to check for TLS. Your cancer team will also give you advice about how much fluid to drink before and during your treatment. They will explain if you need:

- extra fluids through a drip
- medicines such as [rasburicase](#) through a drip, or allopurinol as tablets.

Contact the hospital straight away on the 24-hour number if you have symptoms that could be linked to TLS. This includes:

- feeling very hot (feverish) or having chills, and feeling sick
- pain in your joints or muscles
- feeling breathless or having an irregular heartbeat

- dark or cloudy urine
- feeling confused
- pain or swelling in your tummy
- fits or seizures.

## Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

## Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

## Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

## Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

## Muscle or joint pain

You may get pain in your muscles or joints during this treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

## Skin changes

This treatment may affect your skin. It may cause a rash, which might be itchy.

Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

## Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital.

Do not take any more of this treatment and contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

## Nail changes

You may notice changes to your nails. They may become brittle and break easily. Wear gloves when washing dishes or using detergents to help protect your hands and nails. If the area around your nails becomes inflamed, tell your doctor straight away.

## Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness

- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

## Less common side effects

### Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

### Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

### Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

### Blurred vision

Ibrutinib may cause blurry vision. This treatment may also cause pain, swelling or bleeding in the eye, but this is less common.

Always tell your doctor or nurses if you notice any change in your vision.

## Effects on the nervous system

Ibrutinib can affect the nervous system. You may feel dizzy or unsteady.

Tell your doctor or nurse straight away if you notice this. They may make some changes to your treatment if it becomes a problem for you.

It is important not to drive or operate machinery if you notice these effects.

Very rarely, this treatment can cause a condition called PML (progressive multifocal leukoencephalopathy). Symptoms include:

- difficulty speaking
- difficulty walking
- weakness that gets worse over time
- memory loss
- confusion
- personality or behavioural changes
- changes in your eyesight.

Contact the hospital straight away if you have any of these symptoms. It is important not to drive or operate machinery if you notice these effects.

## Increased risk of stroke

Ibrutinib may increase the risk of a mini stroke or stroke, but this is not common. The main symptoms of stroke can be remembered by the following:

- **Face** – your face is drooping on 1 side, you are unable to smile, or your mouth or an eye may have drooped
- **Arms** – you are unable to lift both arms and keep them there because of numbness or weakness on 1 side of your body
- **Speech** – you have difficulty speaking, your speech is slurred or garbled, or you are confused
- **Time** – it is time to dial 999 immediately if you notice any of these signs or symptoms.

## High blood pressure

Tell your doctor, nurse or pharmacist if you have ever had any problems with your blood pressure. Your nurse will check it regularly during your treatment or you may be able to check it at home. Tell them if you have any headaches, feel dizzy or faint, have blurred vision, chest pain or shortness of breath.

## Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away

- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

## Second cancer

This treatment can increase the risk of developing a [non-melanoma skin cancer](#) years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

## Raised levels of white blood cells

Leukaemia or lymphoma can cause very high levels of white blood cells in the blood. Ibrutinib can also cause this for the first few weeks of treatment. This is normal and not a sign that the cancer is getting worse.

You will have regular blood tests while you are taking ibrutinib to check your levels.

## Other important information

### Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

### Hepatitis B reactivation

If you have had a liver infection called hepatitis B, this treatment can make it active again. Your doctor, nurse or pharmacist will talk to you about this. You may have tests for hepatitis B before, during, and after treatment.

## Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

## Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

## Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix<sup>®</sup> for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

## Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

## Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

## Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

## Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

## About our information

### References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

### Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert health professionals and people living with cancer.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is

affected.

You can read more about how we produce our information [here](#).

## Date reviewed

Reviewed: 01 March 2024 | Next review: 01 March 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

## Get in touch

### How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

### Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



#### 0300 1000 200

Our Support Line is available Monday to Friday, 9:30am-4:30pm



#### Email us

Email the Macmillan Fundraising team if you have any questions.

#### Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



#### Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



#### Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

